Explainable Inventory Decision Support through Computational Intelligence: An Enterprise Framework for FMCG Inventory Management

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Abstract

FMCG distributors must set order quantities and safety stocks for many products that share a procurement budget and warehouse space under uncertain demand and lead times. Optimisation can compute such policies but rarely tells a planner how good a policy is, which resource binds or how much service risk remains. Following design science research, we develop an Explainable Inventory Decision Support Framework (EIDSF) and instantiate it in a working, multi-tenant decision support system. EIDSF combines a joint order-quantity and safety-stock model with expected-shortage cost, all-units discounts and budget, warehouse and service constraints; an FMCG-adapted Artificial Bee Colony algorithm (FMCG-ABC); a Lagrangian lower bound that certifies the optimality gap of any recommended policy; and a decision artifact that reports cost drivers, binding resources, model-implied versus realised service and rule-based recommendations. In a seeded study on synthetic portfolios of 6 to 100 products at an equal evaluation budget, penalty-based genetic algorithm, particle swarm and ABC rarely returned feasible policies (0–9 of 30 runs), whereas constraint-adapted variants always did. FMCG-ABC returned the cheapest policies on the main instance, came within about 0.1% of a certified optimum without discounts and had the lowest median gap for most further instance groups, though not for 25 products. Out of sample, model-feasible policies missed individual service targets for about half of the products; modelling the review period reduced the misses from 27 to 9 of 50 at 3.9% higher cost. All evidence is synthetic.

Keywords: Decision support systems, Design science, Inventory optimisation, Artificial bee colony, Explainable optimisation, Lagrangian relaxation, FMCG

1. Introduction

Inventory management in fast-moving consumer goods (FMCG) distribution is a resource-allocation problem. A distributor carries hundreds of products with uncertain demand and variable supplier lead times; each product has its own service requirement; and all products compete for a limited procurement budget and for space in a few warehouses. Setting each product's order quantity and safety stock independently, for example with the economic order quantity (EOQ) and a service-level safety stock, is transparent

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but ignores these shared resources and can produce a policy that the organisation cannot finance or store (Silver et al., 1998; Axsäter, 2015; Zipkin, 2000).

Inventory theory provides the cost and service relationships and, for structured cases, optimality conditions (Hadley and Whitin, 1963; Rosenblatt and Rothblum, 1990); computational intelligence provides population-based methods such as genetic algorithms (GA), particle swarm optimisation (PSO) and the Artificial Bee Colony (ABC) algorithm for non-convex, constrained objectives (Holland, 1975; Kennedy and Eberhart, 1995; Karaboga and Basturk, 2007; Talbi, 2009). Neither delivers a decision on its own: a metaheuristic returns a policy with no statement of its quality, and an optimal policy that a manager cannot interpret is unlikely to be adopted or properly overridden. DSS research (Shim et al., 2002; Arnott and Pervan, 2014) and explainable AI for operational research (De Bock et al., 2024; Coussement et al., 2024; Olan et al., 2025) stress that models must be embedded in the decision process and explain decisions, constraints and trade-offs.

This paper reports a design science study (Hevner et al., 2004; Peffers et al., 2007; Gregor and Hevner, 2013) that develops and evaluates an Explainable Inventory Decision Support Framework (EIDSF) and instantiates it in a working DSS, the Inventory Intelligence Dashboard (IID). We ask how the multi-item decision can be modelled with all costs, discounts and shared resources jointly (RQ1); how ABC should be adapted to the constrained space and which adaptations matter (RQ2); how it compares with EOQ, GA, PSO and standard ABC at an equal evaluation budget and against a certified optimum (RQ3); how robust the policies are to resources, portfolio size, parameters and out-of-sample demand (RQ4); and how optimisation output can become traceable explanations that preserve managerial authority (RQ5).

The contributions are: *(C1)* a joint order-quantity and safety-stock model with expected-shortage cost, all-units discounts and portfolio budget and warehouse constraints that is convex for every fixed price regime (Proposition 1), yielding a Lagrangian bound that certifies the optimality gap of any policy; *(C2)* FMCG-ABC, with EOQ-informed initialisation, service-as-bound encoding, feasibility repair with feasibility-first ranking and a logarithmic multi-dimensional move, each isolated by ablation; *(C3)* a seeded evaluation with an equal evaluation budget, certified and exact references, non-parametric statistics and out-of-sample replay, including the conditions under which FMCG-ABC was *not* best; and *(C4)* a decision artifact whose every entry is computed from the policy, the model or the replay, instantiated in a multi-tenant DSS with an accountable decision workflow. We do not claim *(C5)* that planners decide better with the artifact; a user evaluation is specified as future work.

The work originates in the first author's MPhil thesis, *Artificial Bee Colony Optimization for FMCG Supply Chains* (CASPAM, Bahauddin Zakariya University); its 30-SKU distributor dataset is not used for results because several units and parameters are unconfirmed (Section 9.3).

2. Related work

Multi-item inventory with shared resources. Reorder-point models protect against demand during an uncertain lead time; under a normal approximation the expected shortage per cycle is the standard deviation of lead-time demand times the standard normal loss function (Silver et al., 1998; Axsäter, 2015; Nahmias, 2009). When items share a budget or storage resource, the optimal policy follows from Lagrange multipliers on the shared resource (Hadley and Whitin, 1963; Rosenblatt and Rothblum, 1990), and Lagrangian relaxation provides bounds when constraints are hard to handle directly (Everett, 1963;

Table 1: Capabilities of the closest works and of this study (✓: stated in the source; –: not stated).

Capability	Rosenblatt and Rothblum (1990)	Pasandideh et al. (2013)	Long et al. (2023)	Tan et al. (2024)	Guo et al. (2025)	This study
Multiple items	✓	✓	–	✓	–	✓
Demand / lead-time uncertainty	–/–	✓/–	✓/–	✓/–	✓/–	✓/✓
Shared budget or capacity constraint	✓	✓	–	✓	–	✓
Quantity discounts	–	✓	–	–	–	✓
Metaheuristic optimisation	–	GA, MA	PSO	GWO	ABC	ABC, GA, PSO
Statistical comparison of methods	–	✓	–	–	–	✓
Certified optimality gap	–	–	–	–	–	✓
Out-of-sample policy evaluation	–	–	–	–	–	✓ (synthetic)
Decision-oriented explanation and rules	–	–	–	–	–	✓

MA: memetic algorithm; GWO: multi-objective grey wolf optimiser. Tan et al. (2024) treat storage space as an objective, not a constraint.

Fisher, 1981). All-units discounts make purchasing cost and budget use discontinuous, which breaks the convexity these methods rely on (Zipkin, 2000).

Computational intelligence. ABC (Karaboga and Basturk, 2007, 2008) has many variants, for example multi-dimensional perturbation (Akay and Karaboga, 2012), and wide supply chain use (Kaya et al., 2022; Guo et al., 2025). For multi-item problems with shared resources, Pasandideh et al. (2013) compare a tuned GA and a memetic algorithm under storage capacity and discounts, and Tan et al. (2024) combine Monte Carlo simulation with a multi-objective grey wolf optimiser under a budget. Constraint handling is often by penalty rather than feasibility-first rules (Deb, 2000), and comparisons often use best runs without equal budgets or reference optima (Derrac et al., 2011).

Decision support and explainability. DSS combine data, models and an interface to support a decision maker in a semi-structured problem (Shim et al., 2002; Power, 2002), and design science has become the dominant DSS research paradigm (Arnott and Pervan, 2014). Explainable AI developed mainly for predictions, with post-hoc attribution such as LIME and SHAP (Ribeiro et al., 2016; Lundberg and Lee, 2017; Barredo Arrieta et al., 2020); for operational research, explanations must address the decision model and its outputs (De Bock et al., 2024; Coussement et al., 2024). In optimisation-based inventory planning the natural explanatory units are cost components, binding resources, the service gap and sensitivities. Table 1 positions the study; the contribution is the integration of these elements in one evaluated artifact, not a new general-purpose metaheuristic.

3. Research method and framework

We followed the design science methodology of Peffers et al. (2007). From the problem and the DSS literature we derived six design requirements: integrate decision data with provenance (DR1); represent costs, uncertainty and shared resources jointly (DR2); return feasible policies of known quality (DR3); report service risk honestly (DR4); explain the decision in management terms (DR5); and preserve managerial authority and traceability (DR6). The evaluation is ex post and artificial in the terms of the FEDS framework (Venable et al., 2016): analytical (convexity, bounds), experimental (equal-budget comparison, ablation, sensitivity, scalability, replay) and by automated testing of the model equations and the DSS.

EIDSF organises the decision process into seven stages (Figure 1): enterprise data (ED), decision modelling (DM), computational intelligence (CI), knowledge extraction (KE), explanation (EX), recommendation (RE) and managerial evaluation (ME). The optimised policy is an intermediate artifact that is analysed, explained and then evaluated by a manager, not the decision itself.

The model and methods are implemented in one Python package used both by the experiments and by the DSS (Section 8.1).

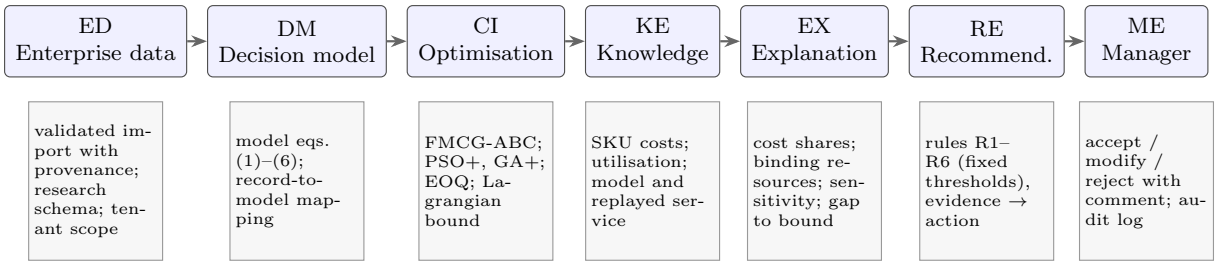


Figure 1: EIDSF stages (top) and their implementation in the Inventory Intelligence Dashboard (bottom).

4. Decision model

We consider N SKUs assigned to warehouses $w = 1, \dots, W$ (I_w the SKUs in warehouse w) over a 365-day year. SKU i has mean daily demand d_i with standard deviation $\sigma_{d,i}$, annual demand $D_i = 365 d_i$, mean lead time L_i (days) with standard deviation $\sigma_{L,i}$, ordering cost K_i, holding cost h_i per unit and year, shortage cost $c_{s,i}$ per unit short, regular price p_i^{reg}, discounted price $p_i^{disc} < p_i^{reg}$ for orders of at least q_i^{disc} units (all-units discount), unit volume v_i and target cycle service level $\alpha_i \geq 0.5$; B is the procurement budget and Cap_w the warehouse capacity. Demand and lead time are independent, lead-time demand is approximated as normal, and shortages are backordered. The decisions are Q_i and SS_i, with $ROP_i = d_i L_i + SS_i$. The standard deviation of lead-time demand and the model-implied service level are

$$\sigma_{DLT,i} = \sqrt{L_i \sigma_{d,i}^2 + d_i^2 \sigma_{L,i}^2}, \qquad z_i = SS_i / \sigma_{DLT,i}, \qquad CSL_i^{\text{model}} = \Phi(z_i). \tag{1}$$

With the standard normal loss function $\mathcal{L}(z) = \phi(z) - z\,[1 - \Phi(z)]$, the expected shortage per cycle is $\sigma_{DLT,i}\mathcal{L}(z_i)$ and the annual cost is

$$TC(Q, SS) = \sum_{i=1}^{N} \Big[\underbrace{K_i \frac{D_i}{Q_i}}_{C_{o,i}} + \underbrace{h_i \Big(\frac{Q_i}{2} + SS_i \Big)}_{C_{h,i}} + \underbrace{c_{s,i} \frac{D_i}{Q_i} \sigma_{DLT,i}\, \mathcal{L}(z_i)}_{C_{s,i}} + \underbrace{D_i\, p_i(Q_i)}_{C_{p,i}} \Big], \tag{2}$$

where $p_i(Q_i) = p_i^{disc}$ if $Q_i \geq q_i^{disc}$ and p_i^{reg} otherwise. Because purchasing dominates TC, all relative gaps are computed on the *controllable cost* $C^{\text{ctrl}} = TC - C_0$ with the constant $C_0 = \sum_i D_i \min(p_i^{reg}, p_i^{disc})$. The model is

$$\min_{Q,SS} \; TC(Q, SS) \quad \text{s.t.} \quad \sum_i p_i(Q_i)\, Q_i \leq B, \tag{3}$$

$$\sum_{i \in I_w} v_i \,(Q_i/2 + SS_i) \leq Cap_w, \qquad w = 1, \dots, W, \tag{4}$$

$$SS_i \geq z_{\alpha_i} \sigma_{DLT,i}, \quad z_{\alpha_i} = \Phi^{-1}(\alpha_i), \tag{5}$$

$$Q_i^{\min} \leq Q_i \leq Q_i^{\max}, \qquad 0 \leq SS_i \leq SS_i^{\max}. \tag{6}$$

Constraint (3) limits the purchase value of one batch of every SKU at the active price (the budget formulation of the thesis); the average-investment form $\sum_i p_i(Q_i)(Q_i/2 + SS_i) \leq B$ is tested as a variant (Section 7.6). A second variant relaxes continuous review by subtracting the expected daily-review undershoot u_i from the protection, $z_i = (SS_i - u_i)/\sigma_{DLT,i}$.

Convexity and a certified lower bound. Let $\bar{\Phi} = 1 - \Phi$ and $R(z) = \bar{\Phi}(z)/\phi(z)$ (Mills' ratio).

Lemma 1. *For every $z \geq 0$, $2\mathcal{L}(z)\phi(z) - \bar{\Phi}(z)^2 > 0$.*

Proof. Dividing by ϕ^2, the claim is $R^2 + 2zR - 2 < 0$, i.e. $R(z) < b(z) := \sqrt{z^2+2} - z$. With $R' = zR - 1$, one finds $b' - (zb - 1) = z(z\sqrt{z^2+2} - z^2 - 1)/\sqrt{z^2+2} \leq 0$ for $z \geq 0$ (equality only at 0), because $(z^2+1)^2 - z^2(z^2+2) = 1$. Hence $\Delta = b - R$ satisfies $\Delta' - z\Delta \leq 0$, so $e^{-z^2/2}\Delta(z)$ is non-increasing on $[0, \infty)$ with limit zero (both R and b are bounded); it is therefore positive, and $\Delta(z) > 0$. □

Proposition 1. *For a fixed price regime of each SKU, model (2)–(6) is a convex program if $\sigma_{DLT,i} > 0$, $c_{s,i} > 0$, $D_i > 0$ and $\alpha_i \in [0.5, 1)$. The same holds for the average-investment budget and the review-period variant.*

Proof. Within a regime, $C_{o,i}$ is convex, $C_{h,i}$ linear, $C_{p,i}$ constant and the constraints linear. For $g(Q,S) = c\,\mathcal{L}(S/\sigma)/Q$ with $c = c_s D\sigma > 0$, using $\mathcal{L}' = -\bar{\Phi}$ and $\mathcal{L}'' = \phi$: $g_{QQ} = 2c\mathcal{L}/Q^3 > 0$, $g_{SS} = c\phi/(\sigma^2 Q) > 0$ and $\det \nabla^2 g = c^2(2\mathcal{L}\phi - \bar{\Phi}^2)/(\sigma^2 Q^4)$, which is positive on the feasible set $z \geq z_{\alpha_i} \geq 0$ by Lemma 1. The variants only add a linear constraint or an affine shift of z. □

Discounts therefore make the problem non-convex only through the choice among 2^N price-regime combinations. Relaxing (3) and (4) with multipliers $\lambda \geq 0$, $\mu \geq 0$ gives

$$g(\lambda,\mu) = \sum_{i=1}^{N} \min_{(Q_i, SS_i) \in X_i} \Big[TC_i + \lambda p_i(Q_i) Q_i + \mu_{w(i)} v_i \big(\tfrac{Q_i}{2} + SS_i\big) \Big] - \lambda B - \sum_w \mu_w Cap_w \leq TC^* \quad (7)$$

by weak duality (Everett, 1963; Fisher, 1981), where X_i collects (5)–(6). The relaxation separates by SKU; each SKU is solved per price regime and the cheaper kept, so the 2^N combinations are never enumerated. For fixed Q_i the optimal safety stock has the closed form $1 - \Phi(z_i^*) = (h_i + \mu_{w(i)} v_i) Q_i / (c_{s,i} D_i)$, clipped to $[z_{\alpha_i}, SS_i^{\max}/\sigma_{DLT,i}]$; by Proposition 1 the partial minimum is convex in Q_i and is found by golden-section search on $\log Q_i$. We maximise g by coordinate ascent; every visited point is a valid bound. Without discounts strong duality holds (Boyd and Vandenberghe, 2004) and the bound is tight; with discounts it remains valid but a duality gap can remain. Either way, the gap to the bound is an upper bound on the true optimality gap, which the DSS reports with every recommendation.

5. FMCG-ABC

All stochastic methods search $x \in [0,1]^{2N}$, which a common decoder maps to (Q, SS), call the same evaluation function and stop after $E_{\max} = 50{,}000$ evaluations. In standard ABC (Karaboga and Basturk, 2007), $SN = 50$ food sources are improved by employed and onlooker bees that perturb one dimension, $v_{jk} = x_{jk} + \varphi_{jk}(x_{jk} - x_{rk})$ with $\varphi_{jk} \sim U(-1,1)$, with greedy selection; a source not improved for 120 trials is replaced by a random scout; onlookers are chosen rank-proportionally. FMCG-ABC adds three groups of adaptations, evaluated cumulatively:

A2 EOQ-informed initialisation. Half of the sources start around the EOQ policy ($\pm 20\%$ in Q, $SS_i = z_{\alpha_i}\sigma_{DLT,i}(1 + U(0, 0.2))$), half uniformly.

Table 2: Synthetic portfolio generator (per SKU unless stated).

Demand	mean LogNormal($\ln 40, 1$) clipped to [2, 600]/day; Gamma series, $CV_i \sim U(0.15, 0.75)$
Lead time	$L_i \sim U\{3, \ldots, 21\}$ days; $\sigma_{L,i} = L_i \cdot U(0.1, 0.3)$
Price; discount	$p_i^{reg} \sim U(80, 1500)$; rate $U(0.02, 0.08)$; $q_i^{disc} = U(1, 2) \times Q_i^{EOQ}$
Costs	$h_i = U(0.18, 0.30)\, p_i^{reg}$; $K_i \sim U(1500, 6000)$; $c_{s,i} = U(0.25, 0.80)\, p_i^{reg}$
Volume; warehouses	$v_i \sim U(0.0008, 0.012)\,\mathrm{m}^3$; $W = 2$
Service target	0.98 / 0.95 / 0.90 for the top 20% / next 30% / last 50% by demand value
Budget; capacity	$B = \tau_B \sum_i p_i(Q_i^{EOQ}) Q_i^{EOQ}$; $Cap_w = \tau_C \sum_{i \in I_w} v_i(Q_i^{EOQ}/2 + SS_i)$

A3 Constraint adaptation. (i) The service constraint is encoded as a lower bound of SS_i. (ii) If (3) or (4) is violated, (Q, SS) is moved towards its lower bounds along a segment whose parameter is found by 30 bisection steps keeping a feasible end, and the repaired policy is written back (Lamarckian repair); if even the lower-bound policy is infeasible, infeasibility is reported. (iii) Sources are ranked feasibility-first (Deb, 2000). Without A3, constraints are handled by the quadratic penalty $F = TC + \rho \sum_j \tilde{V}_j^2$ with normalised violations and $\rho = 10^3\, TC(\mathrm{EOQ})$.

A4 Inventory-specific move (full FMCG-ABC). Order quantities are decoded on a logarithmic scale, so moves are multiplicative and comparable across SKUs, and each dimension is perturbed with probability $MR = 0.1$ (Akay and Karaboga, 2012).

The reported solution is the best feasible policy evaluated; if none is feasible the run is reported as infeasible. *Comparators*: EOQ with static safety stock (all-units-discount EOQ per SKU, ignoring coupling constraints); a real-coded GA (tournament, SBX $p_c = 0.9$, $\eta_c = 15$, polynomial mutation, elitism, population 100) (Deb and Agrawal, 1995); and global-best PSO with constriction coefficients (swarm 100) (Clerc and Kennedy, 2002). GA, PSO and ABC (A1) use the penalty. To separate the search engine from the inventory adaptations, *GA+* and *PSO+* receive exactly the A2–A3 adaptations. All parameters were fixed before the experiments.

6. Evaluation design

The experiments use a documented, seeded generator (Table 2). Daily demand is Gamma distributed (not normal); days 1–730 calibrate d_i and $\sigma_{d,i}$ and days 731–1000 are held out. The budget and capacities are fractions τ_B, τ_C of what the EOQ policy would use, so the coupling constraints bind. No result is based on real company data.

E1 compares EOQ and eight stochastic methods (GA, PSO, ABC, A2, A3, FMCG-ABC, GA+, PSO+) on one 50-SKU instance ($\tau_B = 0.75$, $\tau_C = 0.80$; 30 runs each). *E2* removes the discounts, so the bound is tight and certifies the optimum. *E3* uses five 6-SKU discounted instances whose optimum is obtained by enumerating all 2^6 price regimes, each a convex program, and certified by per-regime bounds. *E4/X3* test five further 50-SKU instances at tight ($\tau = 0.60$) and loose ($\tau = 0.90$) resources; *E5/X2* test $N \in \{25, 50, 75, 100\}$ on three instances per size; *E6/X4* re-optimise under one-at-a-time parameter changes; *X1* gives GA+ and PSO+ equal wall-clock time; *X5* and *X6* use the average-investment budget and the review-period variant; and *E7* replays the median policy of each method on the holdout. In the *replay*, a fixed policy (Q_i, ROP_i)

Table 3: Main comparison on the synthetic 50-SKU portfolio ($N = 50$, $\tau_B = 0.75$, $\tau_C = 0.80$; 30 independent runs per stochastic method, $E_{\max} = 50{,}000$ objective evaluations). Cost statistics are over feasible runs only, on the controllable-cost scale (millions of currency units).

Method	Feasible	U_B (%)	$\max_w U_w$ (%)	Median	IQR	Mean	SD	Δ_{BKS} (%)	Time (s)
EOQ + static SS	0/1	133.3	125.0	15.98	0.00	15.98	0.00	–	0.0
GA	9/30	94.9	99.8	78.52	7.25	78.89	4.83	164.3	2.1
PSO	0/30	85.6	98.9	–	–	–	–	–	0.4
ABC (A1)	1/30	93.5	99.7	80.51	0.00	80.51	–	171.0	14.2
FMCG-ABC (A4)	30/30	99.9	99.9	34.10	2.52	34.31	2.23	14.8	31.6
GA+	30/30	99.9	99.9	40.30	4.51	42.18	5.05	35.7	3.3
PSO+	30/30	100.0	99.9	40.23	2.09	39.17	4.45	35.4	1.7

U_B, U_w: median budget and highest warehouse utilisation of the returned policy (values above 100 indicate violation). Δ_{BKS}: median gap to the best known feasible solution over all runs. Controllable cost $= TC - \sum_i D_i \min(p_i^{reg}, p_i^{disc})$ (a constant shift). Lagrangian lower bound: 25.02; best known feasible: 29.71.

is simulated with daily review and backorders, random lead times and 30 replications, recording realised cycle service level (CSL), fill rate and costs.

We report feasible runs and, over feasible runs, median and interquartile range of C^{ctrl} and the gap $100(TC - TC^*)/(TC^* - C_0)$ to the best known solution (BKS) or the certified optimum. Because runs of different methods are independent samples, FMCG-ABC is compared with each method by two-sided Mann–Whitney U tests (Mann and Whitney, 1947) with Holm correction (Holm, 1979), bootstrap confidence intervals of the median difference and the Vargha–Delaney effect size A_{12} (Vargha and Delaney, 2000), and a Kruskal–Wallis test (Kruskal and Wallis, 1952) across methods; infeasible policies are never compared on cost. All tables are generated by one command from the stored outputs.

7. Results

All results are from synthetic instances (Section 6).

7.1. Feasibility and cost (E1)

The EOQ policy needs 133% of the budget and 125% of the capacity of the more constrained warehouse; its controllable cost (16.0 million) is not attainable, and imposing the shared resources roughly doubles the controllable cost of the best feasible policies (Table 3). The generic penalty-based methods rarely returned an admissible policy (PSO in none of 30 runs, ABC in one, GA in nine), and their feasible policies cost more than twice the BKS; their solutions violated the constraints only slightly in normalised terms, the typical outcome of a quadratic penalty. All constraint-adapted methods were feasible in every run. FMCG-ABC had the lowest median controllable cost (34.10 million, interquartile range 2.52), ahead of PSO+ (40.23), GA+ (40.30) and A3 (54.21), and found the BKS (29.71 million). The Kruskal–Wallis test rejects equal distributions ($H = 83.9$, $p < 10^{-4}$); the medians of FMCG-ABC were lower than those of A3, GA+ and PSO+ by 20.11, 6.20 and 6.12 million (Holm-corrected $p < 10^{-4}$; $A_{12} = 1.00$, 0.95 and 0.81): the advantage over PSO+ is clear but not uniform.

7.2. Distance to the optimum (E1–E3)

The Lagrangian bound of the main instance is 25.02 million; the BKS is 18.7% and the FMCG-ABC median 36.3% above it. With discounts these are upper bounds on the true gaps, which E2 and E3 measure exactly (Table 4). Without discounts (E2) the bound

Table 4: Gaps to a certified optimum: (a) main instance without quantity discounts (convex; optimum certified by the Lagrangian dual, duality gap 0.0000%); (b) five 6-SKU instances with discounts (optimum by enumerating all 2^6 price regimes).

Method	(a) $N = 50$, convex			(b) $N = 6$, discounts		
	Feas. (%)	Median gap	Max gap	Feas. (%)	Median gap	Max gap
GA	10	49.96	49.96	100	11.58	48.12
PSO	0	–	–	100	47.29	131.26
ABC (A1)	0	–	–	100	17.50	63.85
FMCG-ABC (A4)	100	0.10	0.11	100	7.85	47.30
GA+	100	0.10	0.15	100	7.66	47.39
PSO+	100	0.17	0.22	100	7.61	47.29

Gap $= (TC - TC^*)/(TC^* - C_0) \times 100$ on feasible runs, $C_0 = \sum_i D_i \min(p_i^{reg}, p_i^{disc})$. (a) 10 runs per method, $E_{\max}$ as in Table 3; (b) 10 runs per instance, $E_{\max} = 10{,}000$.

Table 5: Ablation of the FMCG-specific adaptations (main instance).

Configuration	Feasible	Median	Δ_{BKS} (%)	Med. viol. (%)	Realised CSL (%)	Fill (%)
ABC (A1)	1/30	80.51	171.0	0.74	97.8	99.59
A2: +EOQ init.	21/30	60.93	105.1	0.00	96.3	99.61
A3: +constraint adapt.	30/30	54.21	82.5	0.00	94.7	99.68
FMCG-ABC (A4)	30/30	34.10	14.8	0.00	92.9	99.34

Median cost over feasible runs (millions; – if none). Median violation: normalised aggregate violation of the returned solution. Realised CSL and fill rate: holdout replay of the median run (Table 6); for A1 the median run is infeasible, so its replay describes an inadmissible policy.

is tight (duality gap 10^{-6}%), and the adapted methods came within 0.10% (FMCG-ABC, GA+) and 0.17% (PSO+) of the certified optimum in median, while penalty-based PSO and ABC returned no feasible policy. On the small discounted instances (E3) the adapted methods had almost identical median gaps (PSO+ 7.6%, GA+ 7.7%, FMCG-ABC 7.9%), with worst runs 47% above the optimum; 135 of 150 runs ended in a suboptimal combination of price regimes, but within the selected regime they were a median 0.15% from its optimum. The hardness lies in the discrete choice of price regimes, not in the continuous part.

7.3. Which adaptation matters? (ablation)

EOQ-informed initialisation (A2) raised the feasible runs from 1 to 21 of 30 and lowered the median feasible cost from 80.5 to 60.9 million; the constraint handling (A3) made every run feasible (54.2 million); and the log-scale multi-dimensional move (A4) produced the largest single improvement, to 34.1 million (Table 5). The same constraint handling also made GA and PSO competitive (GA+, PSO+), so it is not specific to ABC; what distinguished FMCG-ABC is its combination with the perturbation scheme.

7.4. Computing time, robustness and scalability (X1, E4–E6)

Our ABC evaluates candidates one at a time, whereas GA and PSO are vectorised: FMCG-ABC needed about 32 seconds per run against 2 (PSO+) and 3 seconds (GA+). Given equal wall-clock time (10 to 19 times more evaluations), PSO+ and GA+ reached median controllable costs of 40.24 and 38.35 million against 34.00 million for FMCG-ABC; the difference is significant against GA+ ($p < 0.001$, $A_{12} = 1.00$) but not against PSO+ ($p = 0.064$, $A_{12} = 0.75$).

Table 6: Out-of-sample replay of the median run of each method on the 270-day holdout demand (30 lead-time replications; daily review; backorders).

Method	Feasible	Model CSL	Realised CSL [95% CI]	Fill rate	Short. units/yr	SKUs $< \alpha_i$	Ctrl. cost
EOQ + static SS	no	93.1	90.5 [90.2, 90.9]	99.48	7,996	38	17.47
GA	no	97.6	97.3 [97.2, 97.4]	99.31	10,572	11	28.31
PSO	no	87.0	79.4 [79.2, 79.6]	83.05	259,384	15	162.42
ABC (A1)	no	96.7	97.8 [97.7, 97.9]	99.59	6,264	17	25.99
FMCG-ABC (A4)	yes	95.0	92.9 [92.6, 93.1]	99.34	10,052	27	19.09
GA+	yes	96.3	94.8 [94.6, 95.0]	99.62	5,782	25	17.04
PSO+	yes	95.5	92.3 [92.0, 92.6]	99.48	7,943	28	17.21

Model CSL: mean of $\Phi(z_i)$ (eq. 1); mean target $\bar{\alpha} = 93.1\%$. Realised controllable cost excludes purchasing (millions). Demand in the holdout is Gamma-distributed, not normal.

On five further 50-SKU instances, FMCG-ABC had the lowest median gap at tight (4.7% vs. 10.3% for PSO+ and 13.4% for GA+) and loose resources (5.5% vs. 14.1% and 19.1%), although at loose resources PSO+ attained the best solution more often (3 vs. 1 of 25 runs). Over three instances per size, FMCG-ABC had the lowest median gap for $N = 50$, 75 and 100 (4.2%, 5.3% and 4.9%) but not for $N = 25$, where PSO+ led (8.3% vs. 10.9%); standard ABC was feasible only for $N = 25$. Controllable cost rose by 18.5% with 10% less warehouse capacity, by 14.4% with 10% less budget and by 11.4% with a one-point higher service target, and fell by 16.8% with 10% more budget; demand variability moved it by only -3.2% ($\sigma_d \times 0.8$) and $+7.0\%$ ($\sigma_d \times 1.2$). Binding resources and service targets drive the policy cost more than demand variability does.

7.5. Out-of-sample service (E7)

All feasible policies met their targets on average in the model (mean model-implied CSL 95.0–96.4% against a mean target of 93.1%), but on the holdout (Table 6) the realised CSL was 92.9% for FMCG-ABC, 92.3% for PSO+, 94.8% for GA+ and 94.7% for A3, with 25 to 28 of the 50 SKUs below their individual targets, while fill rates stayed above 99.3%; the EOQ policy realised 90.5% with 38 SKUs below target. Replacing Gamma by normal demand did not remove the shortfall (91.7%, 28 SKUs below target), whereas adding the expected daily-review undershoot $(\sigma_{d,i}^2 + d_i^2)/(2d_i)$ to the reorder points raised the realised CSL to 95.3% (17 SKUs below target) but violated the capacity of the second warehouse (105.4%). The continuous-review assumption, not demand non-normality, is the main cause, and the allowance must be optimised within the model rather than added afterwards. Model feasibility is necessary but not sufficient for service.

7.6. Model variants (X5, X6)

With the average-investment budget (X5), all adapted methods remained feasible in every run, standard ABC in none, and FMCG-ABC again had the lowest median controllable cost (53.64 million, against 56.56 for GA+ and 63.22 for PSO+). With the daily-review undershoot built into the model (X6), the re-optimised FMCG-ABC policy cost 35.42 million, 3.9% more than under continuous review, and raised the realised CSL from 92.9% to 96.8%, with 9 instead of 27 SKUs below target, while respecting the capacities; GA+ and PSO+ reached 96.5% and 95.9%. The variant is offered to planners as an alternative policy with its cost and service consequences, not as a replacement.

8. The decision artifact

We demonstrate the decision artifact on the median FMCG-ABC policy of the main instance; it is produced by the same code that the DSS calls.

Table 7: Recommendation rules (condition → evidence → action).

ID	Scope	Condition	Evidence	Action
R1	SKU	$CV_{d,i} > 0.50$ and realised CSL $< \alpha_i$	CV, realised CSL	increase protection, subject to resources
R2	SKU	holding share of controllable cost > 0.50	$C_{h,i}/C_i^{\text{ctrl}}$	review / reduce order quantity
R3	SKU	$L_i > 14$ days	L_i	review supplier lead time
R4	resource	$U_w > 0.95$; status from limit and μ_w	U_w, μ_w	redistribute or reduce exposure
R5	resource	$U_B > 0.95$; status from limit and λ	U_B, λ	delay, split or reprioritise procurement
R6	SKU	realised CSL $< \alpha_i - 0.05$ and $ESE_i > 0.01$	CSL, ESE_i	prioritise service protection

Economic explanation. The artifact decomposes the model cost into its components. Compared with the (infeasible) EOQ policy, the feasible FMCG-ABC policy orders smaller quantities for 33 of the 50 SKUs, which raises ordering cost by 82% and lowers holding cost by 13%, and takes the quantity discount for 17 SKUs; the unconstrained optimum understates the cost of a policy the organisation can finance and store (16.0 against 34.1 million controllable). Holding is the largest controllable cost component for 30 SKUs and ordering for 20; shortage is never the largest, so a manager sees that the policy is shaped by the trade-off between order frequency and stock investment under binding resources.

Constraint and service explanation. The policy uses 99.9% of the budget and 95.2% and 99.8% of the two warehouses. Utilisation alone does not say which limits constrain the policy: the budget and the second warehouse are at their limits with positive multipliers, whereas the first warehouse is highly used but its multiplier is zero ($\lambda = 0.91$; $\mu = (0,\ 17{,}924)$ per m^3). For a 10% change of the budget the multipliers estimate ± 3.73 million, close to the recomputed change of the bound ($+3.91$ and -3.22 million). In the model every SKU meets its target; in the replay 27 SKUs fall short.

Recommendations. Table 7 lists the rules ($ESE_i = ES_i/Q_i$ is the expected fraction of annual demand short); their thresholds were fixed before the experiments and are stored with every artifact. In the first design iteration every rule was attached to SKU records and fired 155 times, including R4 and R5 for all 50 SKUs, which carried no SKU-specific information and flagged a warehouse that does not constrain the policy. In the second, implemented iteration SKU records contain only R1–R3 and R6 (55 firings) and resources are reported once at portfolio level as *constraining*, *high utilisation, not constraining* or *not constraining*, derived from the limit and the multiplier. Table 8 shows the record of the SKU with the most rules fired: it states each fact with the number behind it and the suggested action, and leaves the trade-off (R1 suggests more protection, R2 a smaller order quantity, under a constraining budget) to the manager.

Fidelity. Every number in the artifact is recomputed from (Q, SS) with the model or taken from the replay, and automated tests check that cost components sum to the reported total and that each fired rule's evidence satisfies its condition. This gives a traceable recommendation → evidence → model-output chain. None of the explanations is causal.

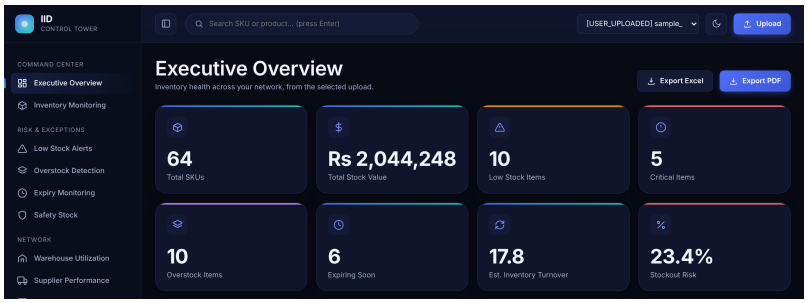
Table 8: SKU-level decision artifact generated by the system for one SKU of the main instance (median FMCG-ABC run). Every entry is computed from the policy, the model or the replay.

Field	Value
SKU	SIM-040 (warehouse 0)
Order quantity Q_i (EOQ baseline)	1,496 (943)
Safety stock SS_i (baseline)	1,030 (1,030)
Reorder point ROP_i	2,952
Price paid / discount taken	1,385.45 / yes
Ordering / holding / shortage cost	51,947 / 723,971 / 73,236
Purchasing cost	48,606,996
Primary controllable driver	holding (85%)
Model / target / realised CSL	0.980 / 0.98 / 0.949
Realised fill rate	0.9887
Budget / warehouse utilisation	99.9% / 95.2%
Rule R1	CV 0.70 $>$ 0.5 and realised CSL 0.949 $<$ target 0.98 $\Rightarrow$ Increase protection inventory, subject to budget and warehouse capacity
Rule R2	holding cost is 85% of controllable cost $\Rightarrow$ Review / reduce the order quantity
Rule R3	lead time 20 d $>$ 14 d $\Rightarrow$ Review the supplier lead-time arrangement

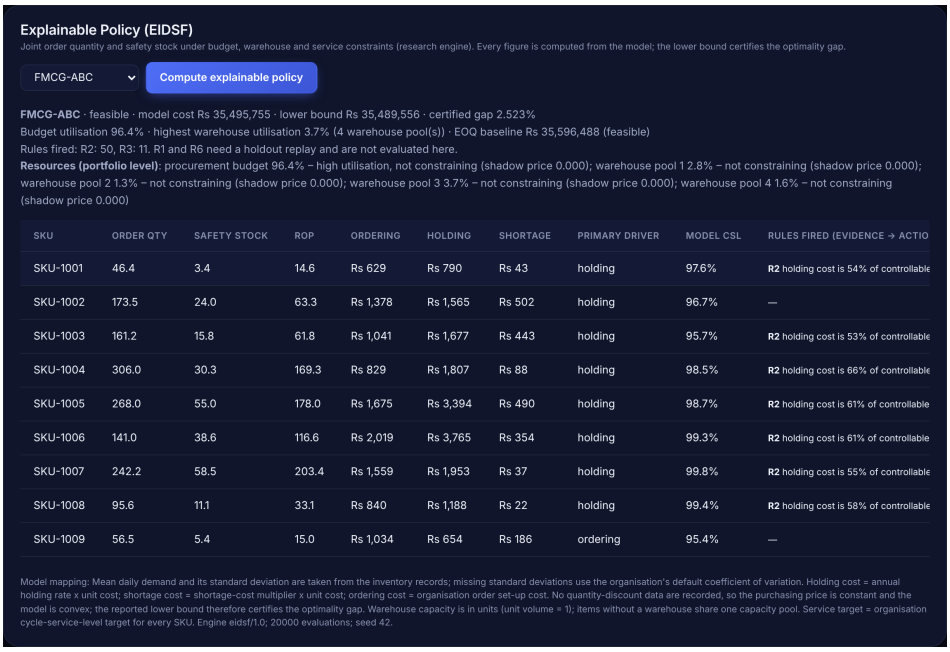
8.1. The Inventory Intelligence Dashboard

The IID (release 2.7.0) is a multi-tenant web application (FastAPI, PostgreSQL, browser frontend) that embeds the artifact in an operational planning workflow (Figure 2). Every record belongs to an organisation; isolation between organisations and the permissions of six roles (owner, administrator, inventory, procurement and warehouse manager, read-only executive) are enforced on the server and tested for every interface operation, and sign-in can require a second factor (time-based one-time passwords). Inventory files are imported only after a validation preview that reports every derived field and records the data provenance (synthetic or customer data). Stock changes after an import (adjustments with a reason code, goods receipts and transfers between warehouses) are posted to a transaction ledger that can be reconciled against the imported quantities. Demand forecasts are produced only from recorded history; products with intermittent demand are classified by the scheme of Syntetos et al. (2005) and forecast with the Syntetos–Boylan approximation (Syntetos and Boylan, 2005) of Croston's method (Croston, 1972), selected by holdout error. Purchase orders follow an approval workflow, and a manager accepts, modifies or rejects each recommendation, with a comment required for modifications and rejections; every decision is stored in an audit log with the user and the evidence. The release passes 299 automated tests, including tests of the model equations, cross-organisation isolation and the browser interface.

The explainable-policy view (Figure 2b) applies the artifact to the organisation's current records: the planner selects a method and obtains the feasibility status, model cost, lower bound and certified gap, the portfolio-level resource status with shadow prices, and one row per SKU with policy, cost components, primary driver, model-implied service and the evidence and action of each fired rule; the mapping from records to model parameters is stated below the table. On the synthetic sample file shown, the budget is 96.4% used but its shadow price is zero, so it is reported as "high utilisation, not constraining"; under the first rule design R5 would have fired for all 64 SKUs. Rules that need a holdout replay (R1, R6) are reported as not evaluated rather than silently omitted. A public demonstration



(a) Executive overview of the selected import.



(b) Explainable-policy view (FMCG-ABC, seed 42, 20,000 evaluations).

Figure 2: The Inventory Intelligence Dashboard, release 2.7.0, on the synthetic 64-SKU sample file distributed with the system (values illustrative and unrelated to the study instances; no discount data, so the bound certifies the gap, here 2.5% of controllable cost). The rules column in (b) is truncated.

instance with the synthetic sample file is available at `https://iid-private.vercel.app` (access for reviewers is provided with the submission).

9. Discussion

9.1. Answers to the research questions

RQ1. The joint model couples SKUs through the budget and the warehouses, and Proposition 1 shows that its only source of non-convexity is the choice of price regimes. The experiments confirm that this is where methods fail: without discounts the adapted methods came within 0.1–0.2% of a certified optimum, whereas on small discounted instances most runs selected a suboptimal regime combination. *RQ2.* The adaptations that matter most are inventory-specific constraint handling, which turned a search that almost never returned a feasible policy into one that always did, and the log-scale multi-dimensional move, which reduced the median gap to the best known solution from 82% to 15%. The constraint handling transfers to GA and PSO, so it is a property of the problem encoding rather than of the bee-colony metaphor. *RQ3.* FMCG-ABC returned the cheapest feasible policies on the main instance at an equal evaluation budget, kept the

lowest median at equal computing time (significantly against GA+ only) and had the lowest median gap for 50 to 100 SKUs and under the average-investment budget, but not for 25 SKUs, and it was practically tied with GA+ and PSO+ on small discounted instances. We claim an advantage on these synthetic instance families, not general superiority; the more robust finding is that generic penalty-based metaheuristics returned inadmissible policies under binding resources. *RQ4.* The policies remained feasible under tighter and looser resources and for larger portfolios; their cost was driven by capacity, budget and service targets more than by demand variability. Out of sample, model-feasible policies missed individual targets for about half of the SKUs, mainly because of the continuous-review assumption. *RQ5.* The artifact turns a policy into checkable statements about cost drivers, binding resources, the gap to a lower bound, the divergence of model-implied and realised service, and the evidence behind each fired rule.

9.2. Implications

Three design principles generalise beyond inventory. First, *feasibility is a first-class output*: a DSS that ranks decisions by a penalised objective can recommend decisions that violate hard business limits; encoding non-negotiable constraints as bounds or repairs, and reporting infeasibility rather than the least-bad candidate, prevents this. Second, *a heuristic recommendation should carry a certificate*: a lower bound on the same instance turns "the algorithm found this" into "this policy is at most $x\%$ more expensive than any feasible policy". Third, *explanations must include what the model does not know*, such as realised next to model-implied service. Unlike the closest multi-item studies (Pasandideh et al., 2013; Tan et al., 2024), the framework reports a bound on solution quality and out-of-sample service; its effect on decision quality, which the explainable-AI literature for operational research calls for (De Bock et al., 2024; Coussement et al., 2024), has not yet been measured. For FMCG planners, the results show that planning SKUs independently understates the controllable cost of a feasible policy by about half on the studied portfolio, that budget and warehouse limits drive the policy, and that service targets set in a continuous-review model should be checked against the review practice actually used.

9.3. Limitations and future research

Data. All results are synthetic. Before the thesis' 30-SKU distributor dataset can be used, the period of its demand standard deviation, the units of lead time and volume, the basis of its holding and shortage costs, the budget, the warehouse capacities and assignment, the service targets and the missing lead-time variability and discount data must be confirmed. *Model and method.* The model assumes independent normal lead-time demand, backorders and average stored volume; the review-period variant still leaves 9 SKUs below target, which a periodic-review formulation or simulation-based service constraints could address. Our ABC implementation is not vectorised, parameters were fixed in advance rather than tuned per method, and with discounts the bound is not tight, which branching over price regimes guided by the dual could improve. *Usefulness to decision makers.* The explanation layer has been evaluated for fidelity, not for its effect on decisions. A feasible next study is a within-subject experiment with 20–30 inventory planners or supply chain students who make replenishment and budget decisions on synthetic portfolios with the optimised policy alone and with the full artifact, in randomised order, measuring the cost and service of their decisions under replay, the detection of seeded errors, decision time, the overrides recorded by the IID workflow and

perceived usefulness. Until then, the DSS contribution is a designed, implemented and technically evaluated artifact, not evidence of better decisions.

10. Conclusion

This paper designed, instantiated and evaluated an explainable decision support framework for multi-item FMCG inventory planning. Its decision model sets order quantities and safety stocks jointly under ordering, holding, expected-shortage and quantity-discounted purchasing costs, a procurement budget, warehouse capacities and service targets, and it is convex within each price regime, which yields a lower bound that certifies the quality of any recommended policy. An ABC algorithm adapted to the problem returned feasible policies in every run and the lowest-cost policies on most synthetic instance groups, while generic penalty-based GA, PSO and ABC rarely returned feasible policies under binding resources; the ablation traces the gain to inventory-specific constraint handling and a log-scale perturbation rather than to the bee-colony search as such. Replaying the policies on held-out demand showed that model feasibility does not guarantee realised service, mainly because of the continuous-review assumption, which the decision artifact reports next to cost drivers, binding resources, the certified gap and rule-based recommendations. The framework runs in a working decision support system in which planners retain and record the final decision. The evidence is synthetic and technical; application to real distributor data and a user study of the explanations are the next steps.

Data and code availability

All results are based on synthetic data fully determined by the documented generator and seeds. The code that generates the data, runs every experiment and produces every table and figure is archived as release v2.4.0 on Zenodo under the MIT licence (Yasir and Javaid, 2026), `https://doi.org/10.5281/zenodo.23529016`. The research engine is unchanged in the dashboard release 2.7.0 described in Section 8.1, whose demonstration instance runs at `https://iid-private.vercel.app`. The company dataset of the first author's thesis is subject to confidentiality restrictions and is not used in any result reported here.

Declaration of competing interest

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Muhammad Yasir: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Imran Javaid:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Anthropic Claude) to help implement and test software, run the experiments and draft and edit text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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